Complementing Speech Interaction Design with Touch for Multi-Robot Systems

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***Abstract*—There has been increasing demand in using multiple robots with human intervention for higher robustness in sectors like agriculture, clinical applications, environment surveillance, military operations, security. The benefits of using multiple robots include increased speed of missions largely due to ability to do parallel tasks and redundancy due to being able to replace a robot with another. However, at the same time, multiple robots substantially increase amount of information exchange with their operator. Therefore, designing interactions between human and multiple robots to achieve their effective cooperation has been a difficult research issue. In this paper, we propose an intuitive complementary user-interface design using speech and touch to control multiple robots simultaneously for a mission. This is achieved using a list of touch complement speech guidelines. We tested the system with six subjects who operated one, two and three robots in a simulation environment built using Unity. The mean performance score for various tasks and perceived cognitive load for each experiment iteration measured using the NASA-TLX questionnaire is then shown to illustrate the benefits of touch complement speech.**

***Index Terms*—Voice User Interface, Complementary User Interface, Multimodal Interface**

I. Introduction

Human multi-robot teaming is often desired to achieve various missions like search and rescue (SAR), surveillance, combat operations, logistics. While robots can perform tedious and dangerous tasks without any error, human intervention is typically still required in performing tasks like critical decision making. For example, robots can perform lane following robustly, there is demand for possible human intervention for giving directional instruction when situations like GPS might is denied. It is generally said that users should have responsibility for global tactics, and robots for local tasks [1]. Therefore, currently the research focus is to improve human-robots cooperation rather than completely autonomous robots.

For a fully autonomous robot, the interaction may consist of high-level supervision and providing way-point navigation. With supervisory control, a human supervises the behavior of an autonomous system and intervenes as necessary. Different attributes that affect the interaction design includes level and behavior of autonomy, nature of information exchange, adaptation, learning, and training of people and the robot, and shape of the task [2]. The type of communication media that helps in the exchange of information includes visual displays, typically presented as graphical user interfaces or augmented reality interfaces, gesture and speech.

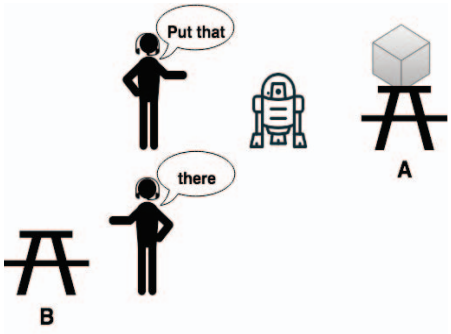


Fig. 1. Human operator commanding the robot to move a box from location A to B in a multi-modal way using gesture to point at locations and speech to specify the task

Speech is one of the most natural way of communication for humans and has gained attention due to its convenience and accessibility. According to statistics, about 75% of human communication in daily life is accomplished through language. Voice is a highly efficient communication media. Therefore, voice interaction has obvious advantages in user experience and hence is proposed to be investigated in greater details. Speech as an interface, have recently gained a lot of attention with various virtual assistants like Siri by Apple and Alexa launched by corporate giants like Google and Sonos. Voice User Interface (VUI) uses speech recognition technology to enable users to interact with technology using just their voices. VUI has several advantages like hands free usage and efficient interactions that are more human in nature than any other form of user interface. But at the same time, VUI also poses difficulties for the user due to several disadvantages like lack of feedback.

Due to the numerous advantages of speech as well as its disadvantages, we investigated the strength and weakness in great details to design a multi-modal touch complement speech interface. Multimodal interaction as shown in Fig. 1 have the ability to leverage the strength and avoid the weaknesses of the individual modality to form coherent and more effective interfaces [3]. Also with more number of robots to control,

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multimodal interfaces helps to lower operator cognitive load. There are three strategies of design multi-modal interfaces, namely: 1) replacement strategy where one modality replaces another in an application that is used on different devices with different input output capabilities; 2) redundancy strategy where different modalities have the ability to convey the same information at the same time and thus have the ability to put additional emphasis on the information that is presented to the user; and 3) complementary strategy where each individual modality provides a best fit for a different part of the task in hand as shown in Fig. 1.

The contribution of the work can be summarized as follows. (i) Complementary and scalable interaction design to control multiple robots using touch to complement speech (ii) Use of touch and voice for different purposes based on their strength (iii) evaluation of the proposed design with a within-subjects study for six participants for three task loads. We studied the mean performance score and the perceived cognitive load using the NASA-TLX questionnaire [4]. The paper will next describle the experiment design, follows by the multi-modal interface design with its analysis. The experimental results is then shown with its analysis followed by conclusion.

II. EXPERIMENT SETUP



Fig. 2. Overview of experiment system architecture : touch-screen monitor showing the simulation environment and Dragon Bluetooth Headset

We used a touch-screen monitor to display the simulation environment as well as to receive touch commands and a Dragon Bluetooth headset to give voice commands as shown in Fig. 2. The primary objective of the task is to handle multiple mobile robotic platforms with certain low level of autonomy.

A. Simulation environment Design

The designed environment represents a remote surveillance mission using multiple robots. The simulation environment consists of multiple semi-autonomous (low level autonomy like line following is available) robots which needs to be teleoperated from their respective start to end locations in an urban environment with buildings, trees, benches, people, etc as shown in Fig. 3. Robots are equipped with top-mounted rotation-enabled surveillance camera to capture targets by precision maneuvering while navigating towards the end locations.

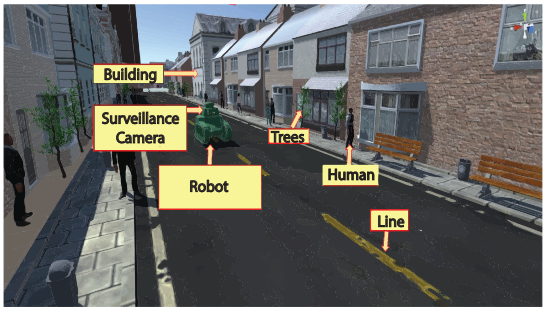


Fig. 3. Urban town simulation environment with different props and the line-following robot

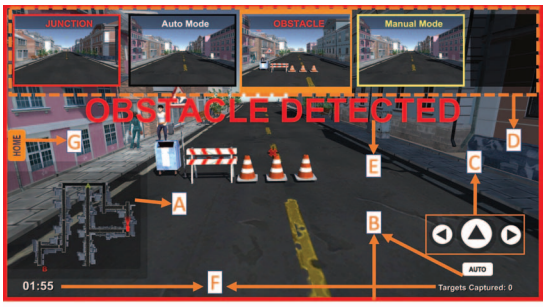


Fig. 4. Graphical User Interface showing surveillance view of the robot as the main view. The user can see the current location of the robot in the environment using A) and check the auto/manual mode using B). To navigate the robot, they utilised touch-screen based control C). D) shows the smaller view of all the robots present in the environment. E) represents the warning of autonomy failure or to seek human intervention. F) shows the time elapsed and target captured since the start of the mission and G) can be used to align the surveillance view in the direction of the robot.

B. Graphical User Interface

We followed an iterative design process to figure out the possible issues from the user's point of view and designed to conform to design principles for interactivity and visualisation to enhance the level of affordance and usability. Overall the GUI is filled with the camera view as if the operator is directly on the robot, and any important/useful information and widgets for control are overlaid on top of this main camera view, mainly arranged on the edges of the screen to minimise blocking of the operators view of the camera view. Because information and widgets are overlaid on the camera view, strategies to reduce visual interference were experimented and designed in. They include: (1) use of bright, strong and primary colors [5]; (2) anti-interference techniques using shadowing behind information/widgets; (3) use of semi-transparent panel behind information/widgets to ensure sufficient foreground-background contrast where necessary. We ensured all widgets (like mode indicators and map label) look similar to make the overall GUI look coherent. Also a good interaction design provides feedback of successful interaction, making it both visible and understandable [6]. On long pressing the smaller robot view, a blue border will be shown as a sign of successful remote selection.

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Various components of the graphical user interface (GUI) shown in Fig. 4 are:

1) Main View - Surveillance view of the camera mounted on top of the robot. When this camera is aligned in direction of robot motion using the home button marked as G) in Fig. 4, it shows the front view.
2) Map View - Orthographic view of the urban environment. It also contains the current location indicator of the robot represented in a small red icon marked as A) in Fig. 4.
3) Navigation Mode Indicators - The robots will be mostly navigating automatically due to the line-following capability. But to simulate the autonomy feature failure, there is a path in the environment, where the users have to drive manually. Users can check the current mode of the robot with the help of the small batch present at bottom right corner of the GUI marked as B) in Fig. 4. Navigation mode is also displayed by the color of the border of the main view of GUI. In Fig. 4, the border color is red, which is indicating a warning. Otherwise, the color of the border is black in auto mode and yellow for the manual mode.
4) Navigation Controls - Since the robot has low autonomy level, on various instance like on a junction, in front of an obstacle, it depends on the human operator to receive commands. So, in all such cases, the operator needs to give commands like Go, Left, Right, Stop present on the GUI marked as C) in Fig. 4 in bottom right corner.
5) Other Robot Views - The smaller robot view present at the top marked as D) in Fig. 4 shows the surveillance views of all the robots, that the participant has to control. In order to identify the current main view from all the small robot views, there is an orange border around that small view whose main view is being displayed.
6) Warning Indication - During the robot navigation, certain warnings were displayed in red color just below the other robot views in the GUI marked as E) in Fig. 4.
7) Time Taken and Targets Captured Indicators - The time that has been elapsed since the start of the mission and the count of captured targets is shown in white color at the bottom of the GUI marked as F) in Fig. 4.

C. Tasks and Controls

For the users, the overall mission goal was to navigate all the robots from a start point to an end point in the environment as soon as possible. And on the way, capture as many photo of people wearing green caps as possible with the help of the surveillance camera of the robot as shown in Fig. 8. Other tasks and controls are described as below:

1) Switching robot views - Since there will be more than one robot for the participant to control, the participant has to keep on shifting his attention from one robot to another in order to multitask. To see any robot view, the user had to just touch on its smaller view shown at the top of the GUI as shown in Fig. 4. On successful selection, the main view of the GUI will change to the selected robot's view and also the orange window will shift to the touched smaller robot view as shown in Fig. 4.
2) Navigation - The users can use the buttons present on the bottom-right corner of the touch-screen as shown in Fig. 4 to give navigation commands to the line-following robots. For example, to start moving in the beginning, choose correct direction to move on the junctions as shown in Fig. 5 and to drive manually on the path without the line as shown in Fig. 6.



Fig. 5. Line-following robots seeking human command for the correct direction on a junction

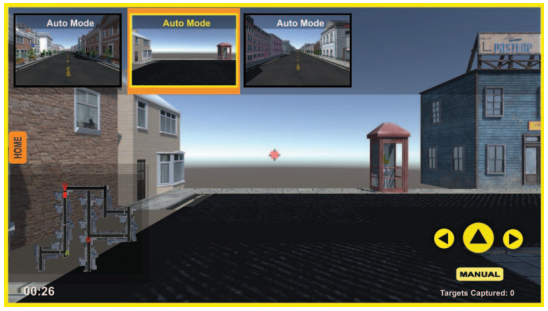


Fig. 6. Robot entering the path with no line to follow. The user has to keep on pressing the navigation control to drive the robot. The yellow border around the GUI indicates manual mode

3) Target Capture - In order to capture targets, the user has to first aim the small red circle present at the center of the robot surveillance camera as shown in Fig. 8 on the green hat. When the red circle is aimed correctly, the user should give the voice command "Capture". The design of target capturing task is analogous to locating victims in an Urban Search and Rescue mission (USAR) where the robots navigate in the disastrous locations and forward information to the rescue teams. On successful capture, the count of target captured present at the bottom right corner of the GUI as shown in Fig. 8 increases by one. Also, there is a visual feedback of the disappearance of hat to confirm successful capture and an audio feedback of the camera click sound.
4) Obstacle Avoidance - On some instances, the autonomous motion of the robot is obstructed by obstacles. In the simulation environment, we have put two kinds

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Fig. 7. Robot stops on detecting a hump which can be overcome by just giving the "Go" command to the robot



Fig. 8. Graphical User Interface showing the surveillance view of the robot as the main view, with 1) small red circle at the center aimed in the green hat, 2) robot encountered an obstacle (road blocks), 3) the user is trying to give the command remotely to the other robot

of obstacle. The first is a simple hump that the robot can just cross over as shown in Fig. 7. This obstacle has been chosen to simulate the false alarm that robot gets in the real Urban Search and Rescue (USAR) mission because of tall grass in some area. The second obstacle consists of road blocks and other objects as shown in Fig. 8 that the robot can not just go over but manually drive around them.

5) Remote control - We also provided the option to remote control a robot. To use this feature, the participant had to long press the smaller view of the robot to which he wants to control remotely as shown in Fig. 8. Before lifting the finger from the screen, the participant should give the voice command to the robot. This feature is useful to avoid missing targets in other robots.
6) Home - Using this feature, users can realign the surveillance camera in the robot heading direction, which is mostly useful in case of junction, obstacle and manual navigation.

D. Experiment Design

The objective of the experiment was to evaluate the user-interface to control multiple robots simultaneously. So, we ran three iterations of experiment with variable task load (1 robot, 2 robots and 3 robots) with six participants. We counter-balanced the order of task load as mentioned in Table I.

TABLE I
ITERATIONS OF COUNTER-BALANCED ORDER OF TASK LOAD.

Participants	Iteration-1	Iteration-2	Iteration-3
Participant 1	2	1	3
Participant 2	2	3	1
Participant 3	3	2	1
Participant 4	3	1	2
Participant 5	1	2	3
Participant 6	1	3	2

We ran the experiment for six participants both male and female. To prevent bias, we ensured that the same experience is given to all participants by keeping the same mediator for all participants. For each participant, we ran a practice mission first to make them familiarize with the various controls and functions. The practice mission simulation environment was similar to the actual mission environment. But there were always three robots to control and the path from start to end point was also shorter. Despite the short path, we tried to include all types of tasks as mentioned in subsection II-C, in the practice mission so that the user has practice of each and every scenario. We allowed the participant to take as many practice trials as they want. Only when they felt confident enough, we presented the actual mission to them.

III. PROPOSED SOLUTION

In this paper, we establish some guidelines based on a comprehensive literature review on the strength and weakness of VUI to create a complementary interaction design using speech and touch. The VUI we used in the experiment consists of a highly accurate and fast voice recognition platform Dragon speech recognition software.

There are several advantages of VUIs over the commonly used button interfaces.

- VUI is a natural interaction [7] in human daily life, which makes speech interfaces perceived as more intuitive.
- VUI is a hands free and eyes free interface that can reduces effort while enable the users to multi-task on another task. For example, one will prefer to use VUIs to give the commands while driving.
- VUIs are always flexible [7] as they allow the users to add as much commands as they want. Use of on-screen button input can only have fixed number of commands and the number of commands is limited by the number of buttons.
- VUI allows users to cut through layers of commands due to its flexibility [8]. For example, a typical mouse will need to click "file" followed by "save as" etc. VUI will enable the user to achieve it within a line of command.

On the other hand, VUI also has several weaknesses:

- VUI is typically weaker in accurate quantification. One example is spatial information [9].
- VUI typically requires a command to start the VUI. For example, "Hey Google" for Google Home devices [10].

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- Accuracy of automatic speech recognition is also a challenge. There are several different culture of words and words that pronounce similarly should be avoid.
- Error rates for natural language processing algorithms is one of the challenges in VUI.
- Difficulty in learning and discovering the voice commands [11], [12], [13] which is often overcome through required training, help, tutorials and documents
- Difficulty in tracking back the information due to lack of feedback [12].

With this list of the strengths and weaknesses of speech and the experiment tasks, we came up with the following assignment of different task to different modalities.

A. Touch as a targeting mechanism

Voice is not effective for spatial commands [9] as it is difficult to exactly illustrate a point through words. For example, it is tough to use words to convey location. On the other hand, this limitation can be overcome with the support of other interface like gesture or gaze. So, we used touch as a targeting mechanism.

1) Target Capture: In order to capture target, the participant had to first aim the surveillance camera on top of the green hat and then give the capture command. Once the correct aiming is achieved using touch, the user should use speech to give "Capture" command.

2) Robot Selection: The command for robot selection has to be very fast and robust in order to make switching from one robot to another very efficient for the participant. So we mainly provided touch control which has lower processing time than speech to select robot. There are two types of robot selection:

- Selecting the robot by changing the main view. To do this, the user has to touch once on the smaller robot view of the robot he want to select and the main view of the GUI will change to the chosen robot view. In such case, all the control commands are sent to the robot which appears in the main view.
- Selecting the robot without changing the main view. To do this, the user has to keep on touching the smaller robot view for 0.5 sec, until a blue border, comes around it. In this case, all the speech commands will be sent to this robot as described in III-B1.

B. Voice for interface traversals and multi-tasking

Voice is especially suitable for interface traversals and multi-tasking because it can help users cut through multiple steps of an interface (a user might say "go back" while looking at Web page, instead of having to go up and hit the back button in the app) as well as discharge hands to engage in other activities [8].

1) Remote Control: Therefore, we used speech commands to remote control a robot. Remote controlling a robot means sending the commands like Go, Left, Right, Stop, Home to a robot without changing the current main view. In order to activate it, the user has to keep on pressing the small robot view for 0.5 sec, until a blue border, comes around it as mentioned in III-A2. Then, before lifting up the finger, the user must give the speech command like "Go" to make the robot move if it stationary, "Stop" to stop the robot if it is moving, "Left" to make the robot turn left and "Right" to make the robot turn right. In order to continue controlling the main view robot, the user has to just lift the finger off the small robot view.

2) Multiple Targets: While touch is very efficient to convey direction, location and other spatial entities, it is not sufficient to send complex commands. For example, if the task would have to capture three different types of target : green cap, red cap, blue cap, in that case, we can easily have three different commands using speech : "Capture Green","Capture Red","Capture Blue", while the same thing when implemented using touch would need three separate touch buttons on the screen, making the user-interface cluttered.

C. Touch for Fine Control

Fine control is another issue with VUI [14]. It is not intuitive to precisely quantify the degree/magnitude through voice. For example, one will typically issue a command as Increase the speed rather than increase the speed by 26.5%, which will increase the error rate.

1) Manual Navigation: In manual navigation, the robot is devoid of the line-following capability and is dependent completely on user commands for navigation. In order to prevent the robot from colliding from the walls and follow the correct path, user has to give fine-grained control for left and right. It is much simpler to keep on pressing a button on the screen than keep on saying "LEFT" which is both exhausting and frustrating.

2) Obstacle Avoidance: On its motion, the robot might encounter obstacles like road blocks, rode cones, etc. To overcome such obstacle, the robot has to be turn manually and moved over a different to cross those obstacles. Since this task also involves continuous communication of commands, thus we provided touch controls to do it.

IV. RESULTS

We conducted a within-subjects study with six participants. Each participant performed experiments for three task loads using the proposed speech and touch based complementary user interface design. As mentioned in subsection II-C, the overall mission objective was to navigate all the robots to the finish point as soon as possible and on the way, capture as many targets as possible. So, participants accuracy in capturing the targets and time taken to complete the task was analyzed as mentioned in Table III. As expected, the accuracy decreases with more number of robots to control and time taken to navigate all the robots from start point to end point increases. But the increase is lower than the linear increase as the number of robots to control increases. We also examined the perceived cognitive load in completing the task. The NASA-TLX questionnaire (6 dimensions: mental/ physical/ temporal demand, performance, effort and frustration) was administered to determine the perceived cognitive effort in

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TABLE II
NASA-TLX SCORE AT VARIOUS TASK LOAD LEVEL, COLUMN-1 REPRESENTS MENTAL DEMAND, COLUMN-2 REPRESENTS PHYSICAL DEMAND AND COLUMN-3 REPRESENTS TEMPORAL DEMAND

Task Load	Mental	Physical	Temporal	Performance	Effort	Frustration	Total NASA	Task Difficulty
1	5.00	4.67	5.33	7.83	7.00	5.67	35.50	3.00
2	9.17	6.50	9.33	7.33	10.00	7.50	49.83	5.17
3	12.50	8.50	11.50	8.50	11.67	9.83	62.50	6.67

using the interface to complete the tasks at different task loads. The total NASA score mentioned in Table II also shows with increase in number of robots, the user's perceived load increases, however, the increase in perceived load is lower than the linear increase.

TABLE III
MEAN PERFORMANCE SCORE AT VARIOUS TASK LOAD LEVEL

Mean Performance Measure	Task Load 1	Task Load 2	Task Load 3
Accuracy	1	0.68	0.51
Time Taken(in sec)	311	383	540

The perceived cognitive effort increased when the number of robots that participants were required to control increased. Due to the small sample size, we used Friedmans Test followed by post-hoc Wilcoxon Signed Ranks (Bonferroni Correction) to compare the NASA scores across the task loads on each dimension as mentioned in Table II. We reduced the type 1 error(that is you found a significant difference when there isn't really a difference) through Bonferroni correction.

V. CONCLUSION

In this paper, we have explored speech and touch as the communication media to create an interface for multi-robot system. Since speech recognition does not have as high accuracy and response speed as keyboards or buttons, whether the interface with speech technology is effective enough is largely dependent on how well the speech interaction is designed [15]. So, after thoroughly understanding the strength and weakness of speech, this paper proposed a complementary interface design using speech and touch in which some of the operations were designed and assigned to speech and others to touch. For example, the target aiming and manual navigation can only be performed by touch, while remote control of the robots can only be performed by speech. We mitigated the recognition error in speech by using the Dragon Speech engine to leverage on Deep Learning technology which adapts to user voice. We provided regular audio and visual feedback for speech commands. To get a valid user feedback, we conducted the experiments for three task load with six participants using the system and recorded their perceived load using NASA-TLX questionnaire as mentioned in Table II. The proposed interface design also opens up avenue for researchers to explore speech as an interface for more difficult tasks like identifying three different types of targets in a mission which is relevant to the real world search and rescue scenario.

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